

Sécurité Azure



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Au Programme

I. Introduction

II. Objectifs du projet

III. Services Azure

IV. Réalisation

V. Conclusion

Programme

Présentation de la société	01
Problématique	02
Conception et réalisation de la solution	03
Conclusion	04

I. Introduction

1. Qu'est-ce que Azure ?
2. Pourquoi la sécurité dans le cloud est-elle essentielle ?

1. Qu'est-ce que Azure ?



2. Pourquoi la sécurité dans le cloud est-elle essentielle ?



II. Objectifs du projet

1. Organisme d'accueil et contexte

2. Objectifs spécifiques du projet



1. Organisme d'accueil et contexte

RFC (Réseaux Formations & Conseil)

RFC



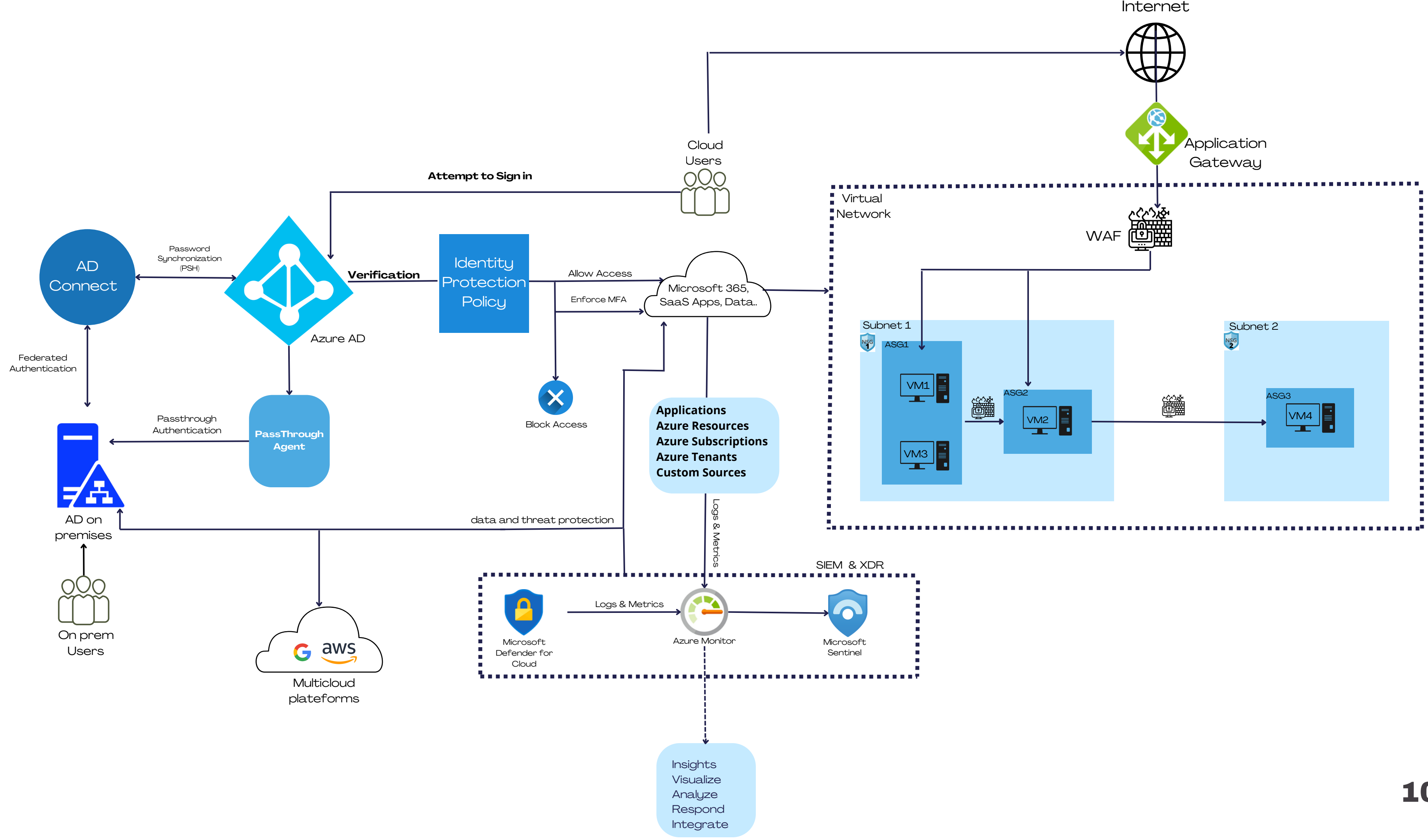
2. Objectifs spécifiques du projet

Sécurité Azure

III. Services Azure

Cette étape est cruciale pour notre projet.

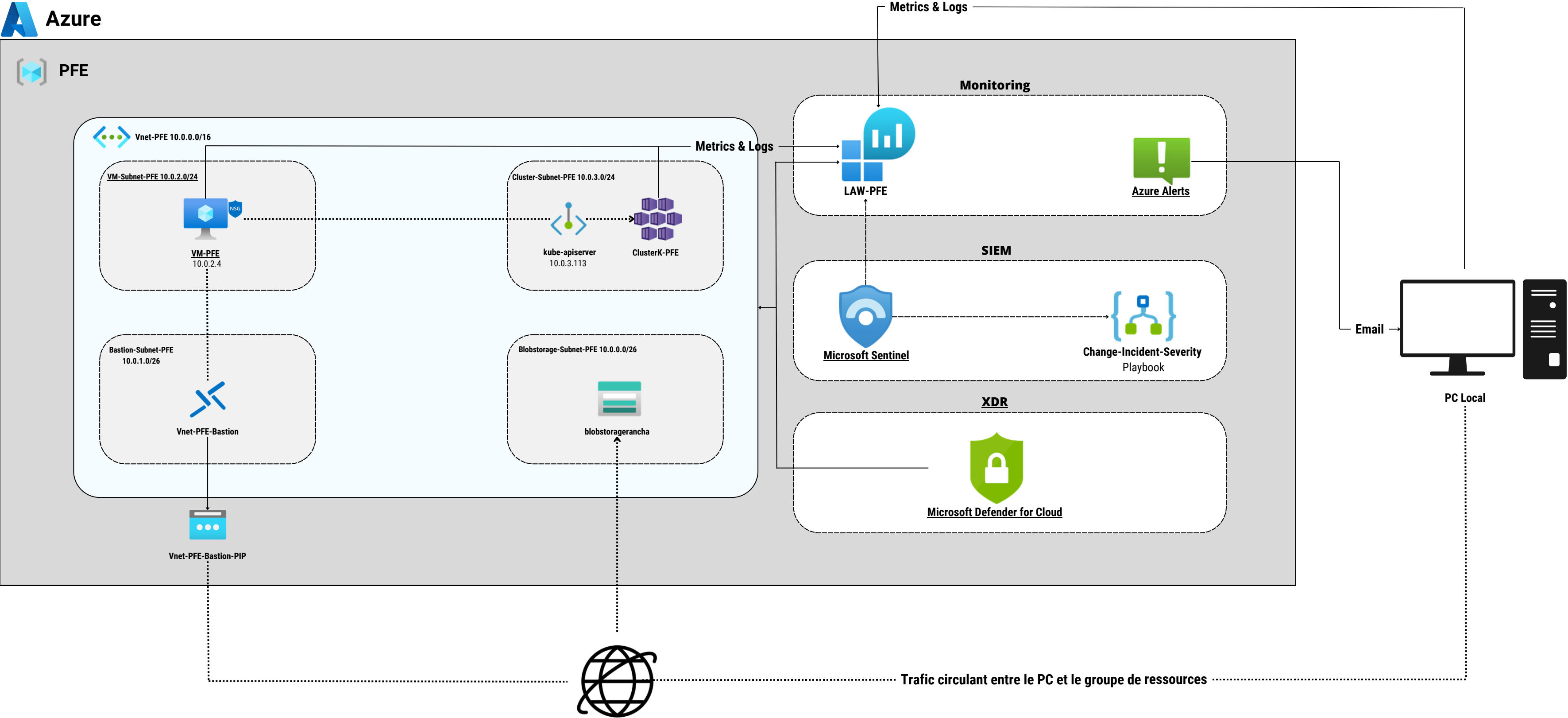




IV. La Réalisation

Nous avons mis en place différents services Azure visant à créer un environnement hautement sécurisé.





Résultat de la création du sous réseau :

Vnet-PFE | Subnets

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Address space

Connected devices

Subnets

+ Subnet

+ Gateway subnet

Refresh

Manage users

Delete

Search subnets

Name ↑↓	IPv4 ↑↓	IPv6 ↑↓	Available IPs ↑↓	Delegated to ↑↓	Security group ↑↓
Blobstorage-Subnet-P...	10.0.0.0/26	-	59	-	-
VM-Subnet-PFE	10.0.2.0/24	-	250	-	-
Cluster-Subnet-PFE	10.0.3.0/24	-	251	-	-
AzureBastionSubnet	10.0.1.0/26	-	59	-	-

13

Résultat de la création du Bastion:

Bastion-PFE

Bastion

Search

<<

Delete Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Sessions

Configuration

Properties

Locks

Monitoring

Metrics

Logs

Diagnostics settings

Connection Troubleshoot

Automation

Tasks (preview)

Export template

Support + troubleshooting

Essentials

Resource group ([move](#))
[PFE](#)

Location
West Europe

Subscription ([move](#))
[Abonnement Enterprise – for Azure Security Internship](#)

Subscription ID
380c6ab0-a5b6-480d-9220-7bccb33d011a

Virtual network/subnet
[Vnet-PFE/AzureBastionSubnet](#)

Public DNS name
bst-83900dd3-f4db-4cf7-8bb5-72157206d231.bastion.azure.com

Public IP address
[Vnet-PFE-ip](#)

Tier
Standard

Provisioning state
Succeeded

Tags ([edit](#))
[Click here to add tags](#)

SessionsTutorials

SessionId	StartTime (UTC)	TargetSubscriptionId	ResourceType	TargetHostName	TargetResourceGrou... U
No results.					



Résultat de la création du compte de stockage :

Home > Resource groups > PFE >

**blobstoragerancha**

Storage account

Search

<<

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Data storage

Containers

File shares

Queues

Tables

Security + networking

Networking

Azure CDN

Access keys

Shared access signature

Encryption

Microsoft Defender for Cloud

Upload

Open in Explorer

Delete

Move

Refresh

Open in mobile

CLI / PS

Feedback

Essentials

JSON View

Resource group (move) : PFE

Location : West Europe

Primary/Secondary Location : Primary: West Europe, Secondary: North Eu

Subscription (move) : Abonnement Enterprise – for Azure Security

Subscription ID : 380c6ab0-a5b6-480d-9220-7bccb33d011a

Disk state : Primary: Available, Secondary: Available

Tags (edit) : Click here to add tags

Properties

Monitoring

Capabilities (7)

Recommendations (0)

Tutorials

Tools + SDKs

**Blob service**

Hierarchical namespace : Disabled

Default access tier : Hot

Blob public access : Enabled

Blob soft delete : Enabled (7 days)

Container soft delete : Enabled (7 days)

Versioning : Disabled

Change feed : Disabled

NFS v3 : Disabled

Allow cross-tenant replication : Enabled

**Security**

Require secure transfer for REST API operations : Enabled

Storage account key access : Enabled

Minimum TLS version : Version 1.2

Infrastructure encryption : Disabled

**Networking**

Allow access from : Selected networks

Number of private endpoint connections : 0

Network routing : Microsoft network routing

Access for trusted Microsoft services : Yes

**File service**

Large file share : Disabled

blobstoragerancha

15

Résultat de la création de l'espace de travail :

Home > PFE >

 **LAW-PFE**

Log Analytics workspace

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Logs

Settings

Tables

Agents

Usage and estimated costs

Data export

Network isolation

Linked storage accounts

Properties

Locks

Classic

Legacy agents management

Legacy activity log connector

Legacy storage account logs

Legacy computer groups

Delete

The Log Analytics agents (MMA.OMS) used to collect logs from virtual machines and servers will no longer be supported from August 31, 2024. Plan to migrate to Azure Monitor Agent before this date. [Learn more about migrating to Azure Monitor Agent](#)

Essentials

Resource group [\(move\)](#)
[pfe](#)

Status
Active

Location
West Europe

Subscription [\(move\)](#)
[Abonnement Enterprise – for Azure Security Internship](#)

Subscription ID
380c6ab0-a5b6-480d-9220-7bccb33d011a

Tags [\(edit\)](#)
[Click here to add tags](#)

Workspace Name
LAW-PFE

Workspace ID
a702657d-b540-41c7-8537-7df79377428e

Pricing tier
Pay-as-you-go

Access control mode
Use resource or workspace permissions

Operational issues
 [OK](#)

JSON View

Get started with Log Analytics

Log Analytics collects data from a variety of sources and uses a powerful query language to give you insights into the operation of your applications and resources. Use Azure Monitor to access the complete set of tools for monitoring all of your Azure resources.

1

Connect a data source

Select one or more data sources to connect to the workspace

[Azure virtual machines \(VMs\)](#)
[Windows and Linux Agents management](#)
[Storage account log](#)
[System Center Operations Manager](#)

2

Configure monitoring solutions

Add monitoring solutions that provide insights for applications and services in your environment

[View solutions](#)

LAW-PFE

16

Configuration de la VM CI\CD :

Résultat de l'Installation de Jenkins et Nexus sur la VM



Jenkins

```
Installing:
jenkins                noarch                2.387.2-1.1                jenkins

Transaction Summary
=====
Install 1 Package

Total download size: 94 M
Installed size: 94 M
Is this ok [y/N]: y
```



Nexus

```
[root@VM-PFE ~]# systemctl daemon-reload
[root@VM-PFE ~]# systemctl enable nexus.service
Created symlink /etc/systemd/system/multi-user.target.wants/nexus.service → /etc/systemd/system/nexus.service.
[root@VM-PFE ~]# systemctl start nexus.service
```

Configuartion de la VM CI\CD :

Résultat de l'Installation d' Azure Cli :

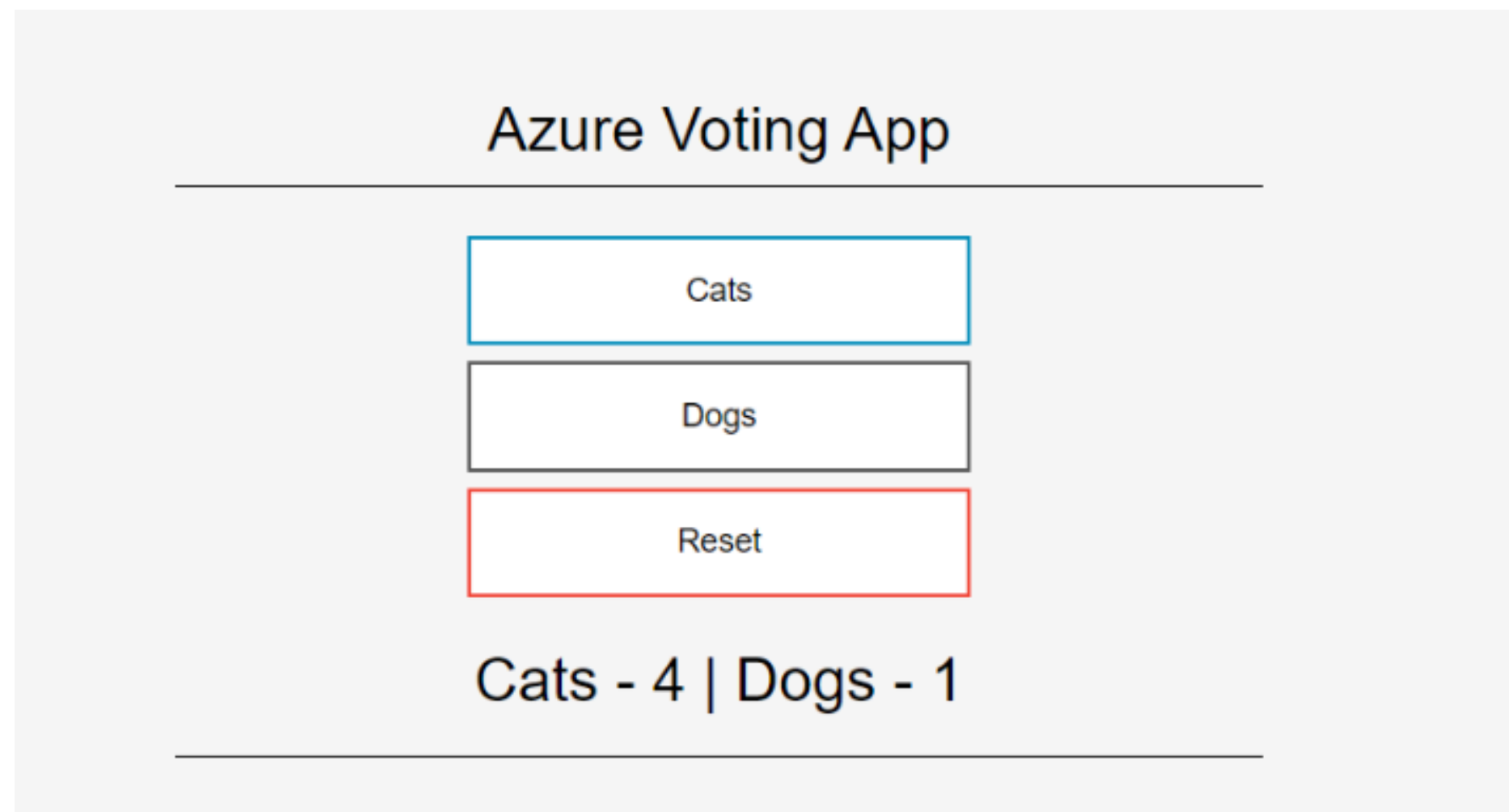
```
[Rancha@VM-PFE ~]$
[Rancha@VM-PFE ~]$ sudo yum install azure-cli
CentOS Linux 8 - AppStream                                19 kB/s | 8.1 kB      00:00
Errors during downloading metadata for repository 'appstream':
- Status code: 404 for http://mirror.centos.org/centos/8/AppStream/x86_64/os/repodata/repomd.xml (IP: 95.216.19.148)
Error: Failed to download metadata for repo 'appstream': Cannot download repomd.xml: Cannot download repodata/repomd.xml: All mirrors were tried
CentOS Linux 8 - BaseOS                                49 kB/s | 8.1 kB      00:00
Errors during downloading metadata for repository 'baseos':
- Status code: 404 for http://mirror.centos.org/centos/8/BaseOS/x86_64/os/repodata/repomd.xml (IP: 149.202.77.145)
Error: Failed to download metadata for repo 'baseos': Cannot download repomd.xml: Cannot download repodata/repomd.xml: All mirrors were tried
Ignoring repositories: appstream, baseos
Last metadata expiration check: 0:00:26 ago on Sun 23 Apr 2023 03:10:19 PM UTC.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
  azure-cli             x86_64           2.38.0-1.el7     azure-cli         50 M
Transaction Summary
=====
Install 1 Package

Total download size: 50 M
Installed size: 709 M
Is this ok [y/N]: █
```

Résultat de l'Installation de kubectl sur la VM :

```
[Rancha@VM-PFE ~]$ sudo curl -LO https://storage.googleapis.com/kubernetes-release/release/v1.22.4/bin/linux/amd64/kubectl
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100 44.7M  100 44.7M    0     0  27.6M      0  0:00:01  0:00:01 --:--:-- 27.6M
```

Résultat de l'installation de l'application sur le cluster via notre VM





Résultat de la configuration des alertes pour le cluster :

Scope

Condition

Actions

Details

Tags

Review + create

Metric alert rule

1 Condition

Terms of use | Privacy statement

Total pricing

Variable

Pricing

Scope

Resource

Abonnement Enterprise – for Azure Security Internship > PFE > ClusterK-PFE

Condition

Signal name

Operator

Aggregation type

Threshold value

Lookback period

Check every

node_memory_working_set_percentage

Greater than

Average

60

5 minutes

1 minute

Actions

Action group name

Cluster-RAM-Action

Contain actions

1 Email

You're now in the Alert-Cluste action group

Boîte de réception x

Microsoft Azure

<azure-noreply@microsoft.com>

À moi

10:19 (il y a 0 minute)

☆

↩

⋮

anglais

>

français

Traduire le message

Désactiver pour : anglais x

Microsoft Azure

You've been added to an Azure Monitor action group

You are now in the Alert-Cluste action group and will receive notifications sent to the group.

View details on Azure Monitor action groups >

20



Résultat de la configuration des alertes pour la VM :

Create an alert rule

ScopeConditionActionsDetailsTagsReview + create

Metric alert rule

1 Condition

Terms of use | Privacy statement

Total pricing

Variable

Pricing

Scope

Resource

Abonnement Enterprise – for Azure Security Internship > PFE > VM-PFE

Condition

Signal name

Operator

Aggregation type

Threshold value

Lookback period

Check every

Available Memory Bytes

Less than

Average

4000000000

5 minutes

1 minute

Actions

Action group name

Mail-Alert-VM-RAM

Contain actions

1 Email

Details

Project details

Subscription

Abonnement Enterprise – for Azure Security Internship

Create

Previous

You've been added to an Azure Monitor action group

You are now in the Mail-Alert-V action group and will receive notifications sent to the group.

[View details on Azure Monitor action groups >](#)

Account information

Subscription ID: 380C6AB0-A5B6-480D-9220-7BCCB33D011A

Resource group name: PFE

Action group name: Mail-Alert-VM-RAM

To unsubscribe from this action group, [click here](#).



[Privacy Statement](#)

Microsoft Corporation, One Microsoft Way, Redmond, WA 98052





Microsoft Defender for Cloud pour la protection du Cluster



ClusterK-PFE

Use Ephemeral OS disk



Abonnement Enterprise – for Azure Security Internship

4/28/2023, 02:57 AM

[Postpone](#) | [Dismiss](#)

az aks nodepool add --name ephemeral --cluster-name ClusterK-PFE --resource-group PFE -s Standard_DS3_v2 --node-osdisk-type Ephemeral

```
[Rancha@VM-PFE ~]$ az aks get-credentials --resource-group PFE --name ClusterK-PFE
Merged "ClusterK-PFE" as current context in /home/Rancha/.kube/config
[Rancha@VM-PFE ~]$ az aks nodepool add --name ephemeral --cluster-name ClusterK-PFE --resource-group PFE -s Standard_DS3_v2 --node-osdisk-type Ephemeral
{
  "availabilityZones": null,
  "count": 3,
  "creationData": null,
  "currentOrchestratorVersion": "1.25.6",
  "enableAutoScaling": false,
  "enableEncryptionAtHost": false,
  "enableFips": false,
  "enableNodePublicIp": false,
  "enableUltraSsd": false,
  "gpuInstanceProfile": null,
  "id": "/subscriptions/380c6ab0-a5b6-480d-9220-7bccb33d011a/resourcegroups/PFE/providers/Microsoft.ContainerService/managedClusters/ClusterK-PFE/agentPools/ephemeral",
  "kubeletConfig": null,
  "kubeletDiskType": "OS",
  "linuxOsConfig": null,
  "maxCount": null,
  "maxPods": 30,
  "minCount": null,
  "mode": "User",
  "name": "ephemeral",
  "nodeImageVersion": "AKSUBuntu-2204gen2containerd-202304.10.0",
  "nodeLabels": null,
  "nodePublicIpPrefixId": null,
  "nodeTaints": null,
  "orchestratorVersion": "1.25.6",
  "osDiskSizeGb": 128,
  "osDiskType": "Ephemeral",
  "osSku": "Ubuntu",
```



Microsoft Defender for Cloud pour la protection de la VM (1)

Home > Virtual machines > VM-PFE | Microsoft Defender for Cloud > Machines should be configured securely > Log Analytics agent should be installed on virtual machines

Machines should be configured securely

View policy definition Open query

Severity: Low

Freshness interval: 24 Hours

Tactics and techniques: Privilege Escalation +2

Locks

Home > VM-PFE

VM-PFE | Locks

Virtual machine

Search

Add Resource group Subscription Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Networking

Connect

Disks

Size

Microsoft Defender for Cloud

Advisor recommendations

Extensions + applications

Continuous delivery

Availability + scaling

Configuration

Identity

Properties

Locks

Add lock

Lock name: Delete-lock

Lock type: Delete

Notes

OK Cancel

Policy

Home > Policy | Definitions > Allowed locations

Allowed locations

Assign policy

Basics Advanced Parameters Remediation Non-compliance messages Review + create

Search by parameter name

Only show parameters that need input or review

Allowed locations

West Europe

west

Germany West Central

Japan West

Jio India West

Norway West

South Africa West

Switzerland West

UK West

West Central US

West Europe

West India

West US

West US (Stage)

West US 2

Review + create Cancel Previous Next

RBAC

Home > Virtual machines > VM-PFE | Access control (IAM)

Add role assignment

Assignment type Role Members Review + assign

Selected role: Virtual Machine Contributor

Assign access to: User, group, or service principal

Members: + Select members

Name: Object ID

No members selected

Description: Optional

Review + assign Previous Next



Microsoft Defender for Cloud pour la protection de la VM (2).

Endpoint protection health issues on machines should be resolved ...

[View policy definition](#) [Open query](#)

Severity
Medium

Freshness interval
 8 Hours

Après Installation de l'agent :

```
configuration property updated  
[Rancha@VM-PFE ~]$ mdatp health --field real_time_protection_enabled  
true
```



Microsoft Defender for Cloud pour la protection du l'espace de travail

Recommendations

Alerts

 Search

▼ More (2)

Severity ↑↓

Description

Status ↑↓

Medium

Microsoft Defender for SQL on machines should be enabled on workspaces

Preview

● Healthy

Medium

Microsoft Defender for servers should be enabled on workspaces

Preview

● Healthy

< Previous

Page

1



of 1

Next >



Microsoft Sentinel (1)

1.Connexion de Sentinel au workspace et au data connector

Home / Microsoft Sentinel /

Add Microsoft Sentinel to a workspace ...

+ Create a new workspace Refresh

Microsoft Sentinel offers a 31-day free trial. See [Microsoft Sentinel pricing](#) for more details.

Filter by name...

Workspace ↑↓

Location ↑↓

ResourceGroup

LAW-PFE

westeurope

pfe

workspace

eastus

pfe2

Home / Microsoft Sentinel / Microsoft Sentinel | Data connectors / Azure Activity /

Configure Azure Activity logs to stream to specified Log Analytics workspace

Assign policy

Basics Advanced Parameters Remediation Non-compliance messages Review + create

Basics

Scope

Abonnement Enterprise – for Azure Security Internship/PFE

Exclusions

--

Policy definition

Configure Azure Activity logs to stream to specified Log Analytics workspa...

Assignment name

Configure Azure Activity logs to stream to specified Log Analytics workspa...

Description

--

Policy enforcement

Enabled

Assigned by

mami

Advanced

Resource selectors

No selectors associated with this assignment.

Overrides

No overrides associated with this assignment.

Parameters

logAnalytics

/subscriptions/380c6ab0-a5b6-480d-9220-7bccb33d011a/resourcegroups...

Remediation

Create

Cancel

Previous

Next



Microsoft Sentinel (2)

2. Création et gestion des playbooks

When a response to an Azure Sentinel alert is triggered (Preview) ...

No additional information is needed for this step. You will be able to use the outputs in subsequent steps.

Connected to ranim.gouider@infratunisia.onmicrosoft.com. [Change connection.](#)

↓

Alert - Get incident (Preview) ...

* Specify subscription id

* Specify resource group

* Specify workspace Id

* Specify alert id

Connected to ranim.gouider@infratunisia.onmicrosoft.com. [Change connection.](#)

↓

Alert - Get accounts (Preview) ...

* Entities list

Connected to ranim.gouider@infratunisia.onmicrosoft.com. [Change connection.](#)

+

↓

Change incident severity 2 (Preview) ...

* Specify subscription id

* Specify resource group

* Specify workspace Id

* Identifier

* Specify alert / incident

* Specify severity

Connected to ranim.gouider@infratunisia.onmicrosoft.com. [Change connection.](#)

[+ New step](#)



Microsoft Sentinel (3)

3. Configuration des règles de détection

 **3**
Active rules

 More content at
Content hub

Rules by severity

High (1) Medium (2) Low (0) Informational (0)

Active rules

Rule templates

Anomalies

Add filter

☐	Severity	Name	Rule type	Status	Tactics	Techniques	Source name
☐	Medium	Delete-Cluster-Rule	 Scheduled	 Enabled	 Initial Access		Custom Content
☐	Medium	Delete-VM-Rule	 Scheduled	 Enabled	 Initial Access		Custom Content



V. Conclusion

En conclusion, notre projet SFE a renforcé la sécurité des infrastructures cloud dans Azure

Merci !

